

NAIVE BAYES

The Simplest Explanation

Naive Bayes is a classification algorithm.

It looks at previous examples, checks the information about a new case, and predicts the most likely group.

One easy idea

Imagine that a bank has records of customers who **repaid** their loans and customers who **defaulted**. A new customer arrives. The algorithm compares the new customer with the old records and decides which group is more likely.

Previous data + New customer information = Most likely prediction

Simple language - Banking examples - Main Naive Bayes types

1. How Does Naive Bayes Work?

Suppose a bank wants to predict whether a customer will **repay a loan** or **default**.

The bank already has information about previous customers.

Previous customer information	What the bank noticed
Good credit score	Many customers repaid
Low credit score	More customers defaulted
Low debt	Many customers repaid
High debt	More customers defaulted

Now consider a new customer

The new customer has:

- A low credit score
- High debt

Naive Bayes checks how often these characteristics appeared in the Repay group and the Default group.

If low credit score and high debt were more common among customers who defaulted, the model predicts **Default**.

That is the basic idea. The algorithm uses probability, but you do not need complicated mathematics to understand its decision.

2. Bayes' Theorem in Simple Words

Bayes' theorem means:

Start with what you already know. Then update your thinking when you receive new information.

Simple banking example

A bank knows that, normally, **10 out of 100 customers default**. Before checking a new customer, the bank starts with a 10% chance of default.

The bank then learns that the new customer has a **low credit score**. In previous records, low credit scores were common among customers who defaulted.

Before seeing the credit score:

Chance of default = 10%

After seeing the low credit score:

Updated chance of default = 30%

The bank changed its estimate because it received new information. This updating process is the main idea behind Bayes' theorem.

Four useful words

Word	Simplest meaning
Prior	What we believed before seeing new information
Evidence	The new information we received
Posterior	Our updated belief after seeing the information
Prediction	The final class selected by the model

Easy sequence: Starting belief -> New information -> Updated probability -> Prediction

3. Why Is It Called 'Naive'?

The model looks at each piece of information separately.

For a loan customer, it may examine:

Information	What it may indicate
Credit score	Good or poor repayment history
Income	Ability to make payments
Debt	Existing financial burden
Employment status	Stability of income

The algorithm makes a simple assumption:

It assumes that each feature gives separate evidence about the result.

In real life, features may be connected. For example, income and loan amount can be related. Therefore, the assumption is called **naive**.

Even with this simple assumption, Naive Bayes often works well and is very fast.

Second banking example: Fraud detection

A new card transaction is made in another country, at midnight, and for an unusually high amount. If these characteristics appeared frequently in earlier fraud cases, Naive Bayes gives the Fraud class a higher probability.

Genuine probability: 35%

Fraud probability: 65%

Prediction: Fraud - the bank can request verification.

4. Main Types of Naive Bayes

The different types are used for different kinds of input data.

Type	Use it when the inputs are...	Simple banking example
Gaussian	Numerical measurements	Age, income, balance, loan amount
Bernoulli	Yes/No or 1/0 values	Missed payment: Yes or No
Multinomial	Counts, especially word counts	Words in complaint messages
Categorical	Named categories	Account type or employment type
Complement	Text data with unequal classes	Rare fraud complaint detection

Easy way to remember:

Numbers -> Gaussian

Yes/No -> Bernoulli

Counts or words -> Multinomial

Categories -> Categorical

Imbalanced text -> Complement

Important note

You do not select a type because one is always better. You select it according to the kind of data you have.

5. Gaussian and Bernoulli Naive Bayes

Gaussian Naive Bayes

Use Gaussian Naive Bayes when the input columns contain **numerical values**.

Numerical column	Example value
Age	42 years
Monthly income	INR 75,000
Credit score	680
Loan amount	INR 800,000
Account balance	INR 120,000

The model learns what numerical values are common in each class. For example, it learns the usual income and credit-score patterns for customers who repaid and customers who defaulted.

Banking use case: Predict loan default using age, income, credit score, balance and loan amount.

Bernoulli Naive Bayes

Use Bernoulli Naive Bayes when every input is mainly **Yes/No, True/False, or 1/0**.

Examples: Has the customer missed a payment? Is the transaction overseas? Is the device new? Is the account frozen?

Banking use case: Predict whether a transaction is fraudulent using several Yes/No warning signals.

6. Multinomial, Categorical and Complement

Multinomial Naive Bayes

Use it when the inputs are **counts**. It is very popular for text because it can count how many times words appear.

Example: A complaint contains the words 'charged' three times and 'refund' twice. The model uses these word counts to classify the complaint.

Categorical Naive Bayes

Use it when the inputs are **categories** rather than measurements.

Example: Account type = Savings, Employment type = Salaried, Risk band = Medium, Region = South.

Complement Naive Bayes

Complement Naive Bayes is mainly useful for **text classification when some classes have far fewer records than others**.

Example: A bank has 9,500 normal service messages but only 500 fraud-related messages. Complement Naive Bayes can be useful for identifying the smaller fraud-message class.

Simple difference

If your data looks like this...	Try this type
45, 680, 75000, 800000	Gaussian
Yes, No, Yes, Yes	Bernoulli
Word counts	Multinomial
Savings, Salaried, Medium	Categorical
Unequal text classes	Complement

7. Final Summary

Naive Bayes makes a prediction by checking which class is more likely based on previous data.

The complete process

Step	What happens
1	The model studies previous records
2	It learns what information is common in each class
3	It receives a new record
4	It calculates the probability of every class
5	It selects the class with the highest probability

30-second classroom explanation

Naive Bayes is a classification algorithm. It studies previous examples and calculates which class is most likely for a new case. It is called naive because it treats each input feature as separate evidence. Gaussian is used for numerical data, Bernoulli for Yes/No data, Multinomial for counts or text, Categorical for categories, and Complement for imbalanced text data.

Key points to remember

- Bayes' theorem updates a probability after receiving new information.
- Naive Bayes predicts the class with the highest probability.
- It is simple, fast and useful as a starting model.
- The correct variant depends on the type of input data.
- A prediction is a probability-based estimate, not a guarantee.

Shortest definition:

Naive Bayes uses past data and probability to predict the most likely category.